HERMES

HERMES

High-Frequency Emergency and Rural Multimedia Exchange System

Email, files, messages and voice over HF radio, where there is no internet

Rhizomatica • **free software** • **November 2026**

github.com/Rhizomatica • hermes.radio



Overview

- A telecommunication system over HF radio for communities beyond the internet and the phone network
- People use it as they use the internet: email, webmail and Delta Chat, over the station's WiFi
- The station carries it over HF to a gateway with an internet uplink
- Also voice, digital and encrypted, broadcast files, and positions with an emergency call
- Free software from Rhizomatica, supported by ARDC grants since 2021; hands-on training so a community runs its own station



The station's web interface: voice, messages, e-mail, the radio, schedules, GPS

Screenshot: hermes-gui

Where HERMES runs









Ecuadorian Amazon

Waorani Indigenous communities: nearly 90 radios, the largest deployment; some sites days away on foot or by boat. Older residents keep in touch with family in other villages

Bangladesh

Fishing communities, HERMES on their fishing boats, in a conservation project

Central African Rep.

Early warning systems

Run by the community

Rhizomatica trains a small group of community members, on site, to install, operate and maintain the station without constant outside support

Source: ARDC, 6 November 2026. Photos: Rhizomatica (Ecuadorian Amazon; training for fishing boats in Bangladesh)

In the field



Getting there: days on foot or by boat

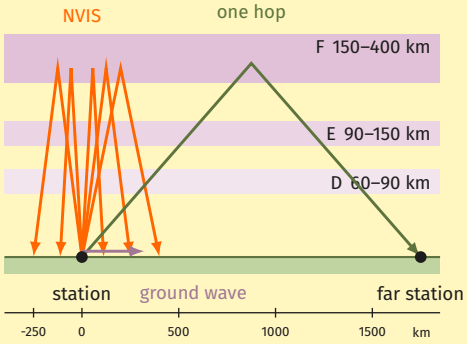


Power: a solar panel and a battery

The station is installed with the community, which then runs it: the radio, the antenna, the power and the computer

Photos: Rhizomatica, Ecuadorian Amazon

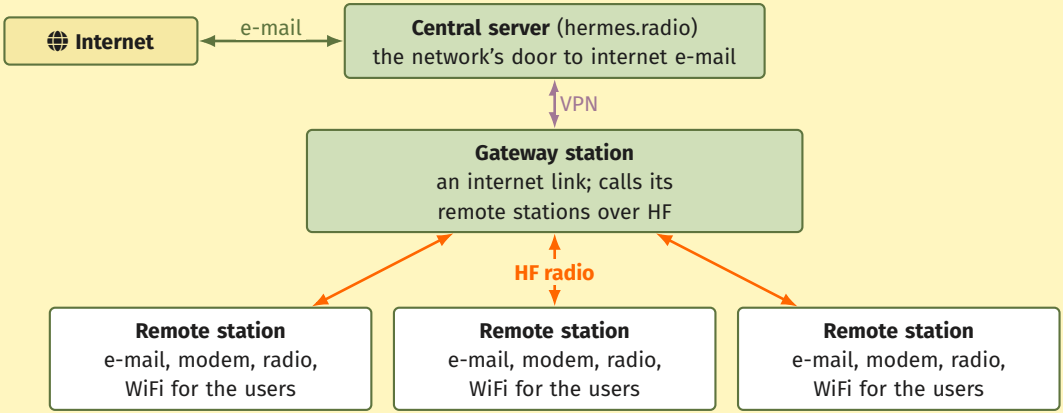
HF: the ionosphere as the relay



- HF is 3 to 30 MHz. The ionosphere, ionised by the sun, bends it back to Earth: no towers, satellites or cables between stations
- NVIS (near vertical incidence skywave): sent almost straight up, on about 2 to 10 MHz, it comes down all around, out to some 400 km, over mountains and forest
- At a lower angle, one hop reaches 1000 to 4000 km
- It changes over the day: the D layer absorbs low frequencies in daylight and fades at night, and the usable frequencies fall after dark, so stations change frequency through the day
- Fading and multipath: the modem adapts its speed, and repeats only what was lost

Distances to scale; heights not

How a network fits together



The central server queues mail for each gateway. The gateway collects it every few minutes, and calls its remote stations over HF on a schedule.

Source: hermes-installer documentation

A station

- A small computer: a Raspberry Pi
- An SSB transceiver: the sBitx or zBitx, or any radio a computer can control (IC-7100, IC-7300, FT-710, ...)
- An antenna, solar power, a WiFi access point; optionally GPS
- A mail server, a store-and-forward queue, the Mercury modem, the radio software and a web interface
- One installer image for every station





Above: a station on a community wall.
Left: the sBitx, with the HERMES interface on its screen

Photos: Rhizomatica

Data over HF: store and forward, and broadcast

- Store and forward: mail and files wait at the station and go at the next call, so nothing needs the link up all the time
- UUCP carries them by default: robust, and it resumes interrupted transfers. NNCP, encrypted and authenticated, is the alternative
- The link is Mercury, with ARQ (automatic repeat request): a file arrives intact, or the sender is told it did not

Mercury, the modem

Free software since May 2026, replacing VARA, a proprietary modem with a paid licence per radio. It adapts its speed to the channel, from a robust mode that works below the noise (−10 dB SNR) up to 3.1 kbit/s. It has its own presentation.

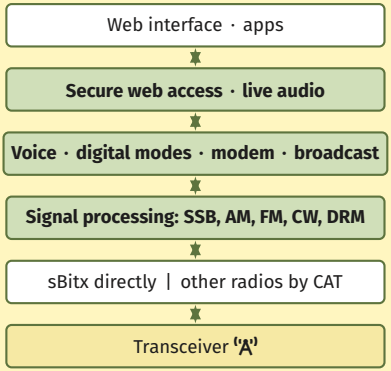
Broadcast: one to many

Files go out under a fountain code (RaptorQ): enough of *any* frames rebuild the file, whenever a receiver tuned in. No return path needed: bulletins and updates to a whole region.

Source: hermes-installer documentation; Mercury's presentation; hermes-broadcast

The radio daemon

- One program runs the station's radio: control, signal processing, voice, digital modes and the modem
- It drives the sBitx and zBitx hardware itself, or any radio with a computer control (CAT) port
- The Mercury modem runs inside it: no audio cables or sound loopback between modem and radio
- The web interface, the apps and the radio's live audio come through one secure web address, with users and roles



Source: hermes-radio-daemon

Digital voice and encryption

Two digital voice codecs

- RADE V2 (FreeDV): neural-network digital voice, made for weak and fading HF channels
- D-STAR: compatible with D-STAR radios
- One switch between analogue and digital voice

Voice encryption (opt-in)

- One 256-bit key shared by the network's stations
- D-STAR: the voice encrypted with ChaCha20
- RADE V2: the key scrambles the modem signal itself; a late listener locks on in about a second

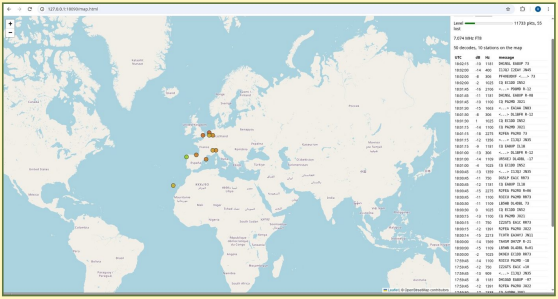
On air, October 2026

Both codecs, encrypted and clear, with the right key, the wrong key and none, and a listener joining mid-over: all passed. Listeners heard no difference between encrypted and clear speech.

Source: hermes-radio-daemon

Digital text modes and reception reports

- FT8: receive and send, decoding weak signals under stronger ones
- JS8: keyboard-to-keyboard messages, at all four speeds
- CW and RTTY skimmers: every station in the passband at once
- Stations heard reported to PSKReporter, at the station's GPS position
- DRM: digital broadcast, received and sent



The station's map: FT8 stations heard, placed by their locators

Source: hermes-radio-daemon

Positions over the air

Position beacons

- A station sends its position on a schedule: from its GPS, or a fixed position typed in
- Every station heard is listed with its last position, and drawn on a map

SOS

An emergency code set by a user goes out in every beacon, more often, until the emergency is declared over

Tracks

- The station's own track, and every other station's from its beacons
- A point every few seconds while moving; one every 5 minutes while still
- A trip is a stretch of time on a track, shown on the map

Source: hermes-radio-daemon

Testing and platforms

Testing

- Automated tests on every change, on PCs and on the ARM boards the stations use
- Each part alone, the whole program on a simulated radio, and checks for memory and threading errors
- HF channels simulated with fading and noise
- On air: sBitx, IC-7100 and FT-710 stations

Platforms

- Raspberry Pi 4 and 5, from the HERMES package repository; Mercury even on a Pi Zero, with the zBitx
- A phone as the station: the radio software and Mercury on Sailfish OS and a PinePhone Pro, the radio on USB, the phone's own GPS
- Mercury also on Android and iOS phones, macOS and Windows; iOS gives apps no USB access

Source: hermes-radio-daemon; Mercury

2026, and what comes next

90 radios in the Waorani communities

12 stations in a network installed in November

580 commits to the radio software since April

1468 commits to Mercury since its rewrite

⇄ Messages and files

One way to send mail and files, whichever link a network uses; an SOS message to chosen contacts

Maps and trips

A station's journeys, drawn on the map

Sharing the frequency

The stations of a network taking turns on one channel; optional encryption in Mercury

Other networks

Reticulum over HF, done; Meshtastic, to reach LoRa networks, in progress

Source: git history of hermes-radio-daemon and Mercury; ARDC, 6 November 2026

HERMES

THANK YOU

github.com/Rhizomatica

hermes.radio • rhizomatica.org/hermes

lists.riseup.net/www/info/hermes-general

HERMES is free software.

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